

Skip List – Insert Element

–Program Analysis

–Insertion(4) [rlevel>level]

But ,what happens,when ,we increase rand() value i.e. 3 > lst → level: 2, then what happens? Let begin this from previous document observation(from middle):

```
if (current == NULL || current->key != key) {...}
```

*current = e216 ≠ NULL (false)OR **current → key = 30 ≠ key = 25(True)**,
hence if block executes.*

```
int rlevel = randomLevel();
```

rlevel = randomlevel() = 3 (say)

```
if (rlevel > lst->level) {  
    for (int i = lst->level + 1; i <= rlevel; i++) {  
        update[i] = lst->header;  
    }  
    lst->level = rlevel;  
}
```

*if **rlevel = 3 > lst → level = 2 is True** [**Condition is True**], if block executes:*

for int i = lst → level + 1 = 2 + 1 = 3; i ≤ rlevel = 3:

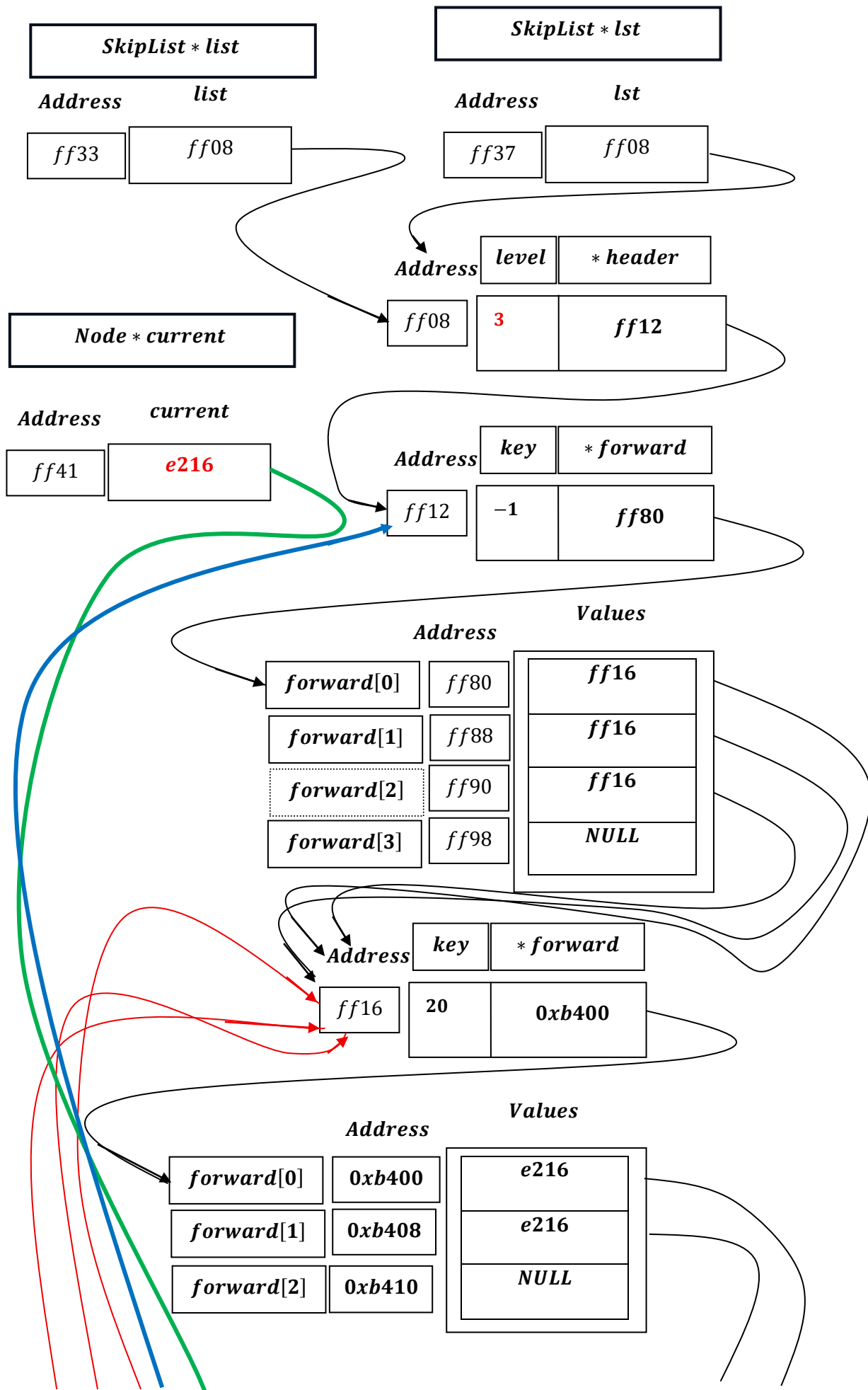
update[3] = lst → header = ff12

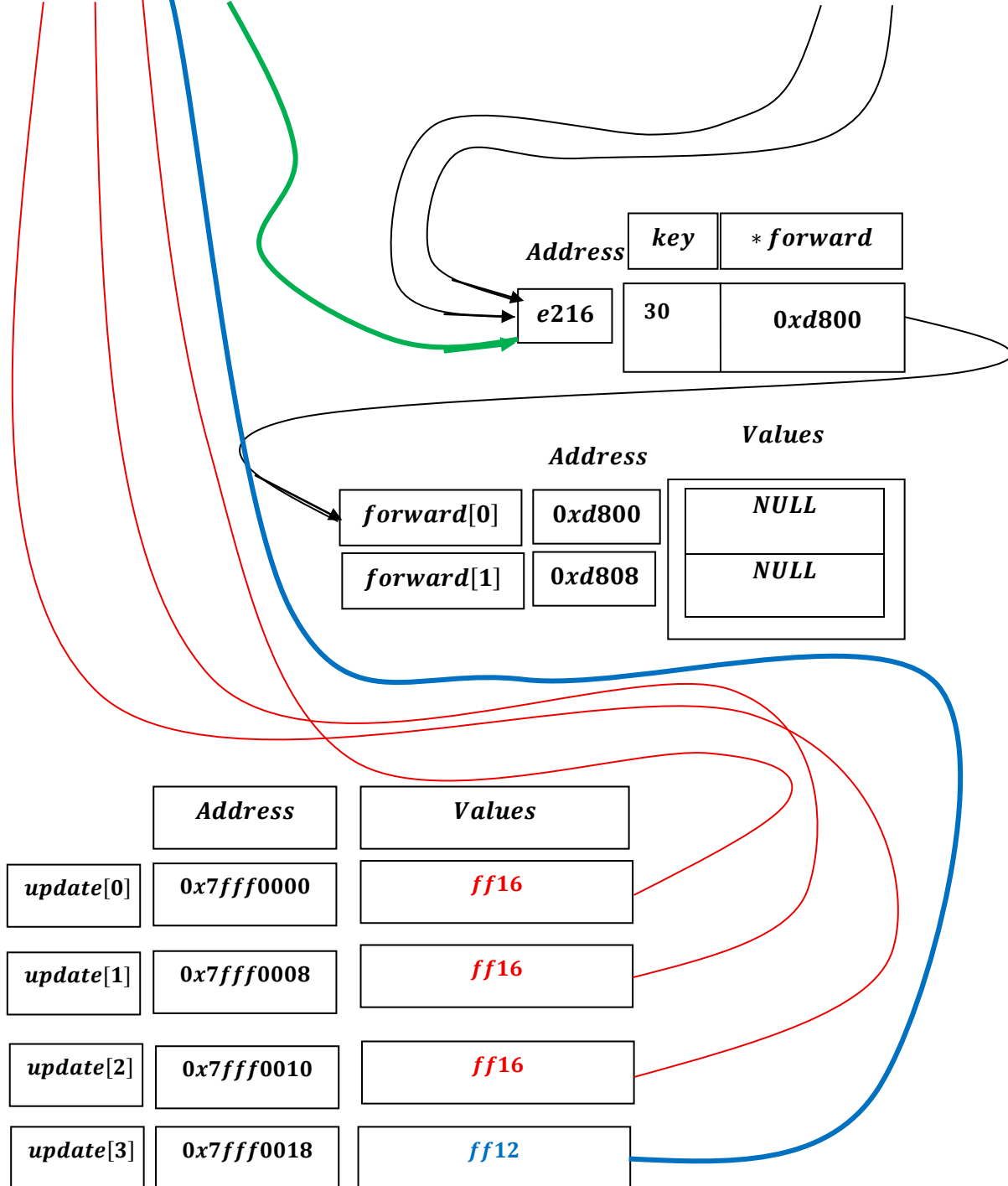
i = i + 1 = 3 + 1 = 4

for int i = 4; i ≤ rlevel = 3 $\left[\begin{array}{c} \textit{Condition is} \\ \textit{false} \end{array} \right]$ *loop exists.*

lst → *level* = *rlevel* = 3.

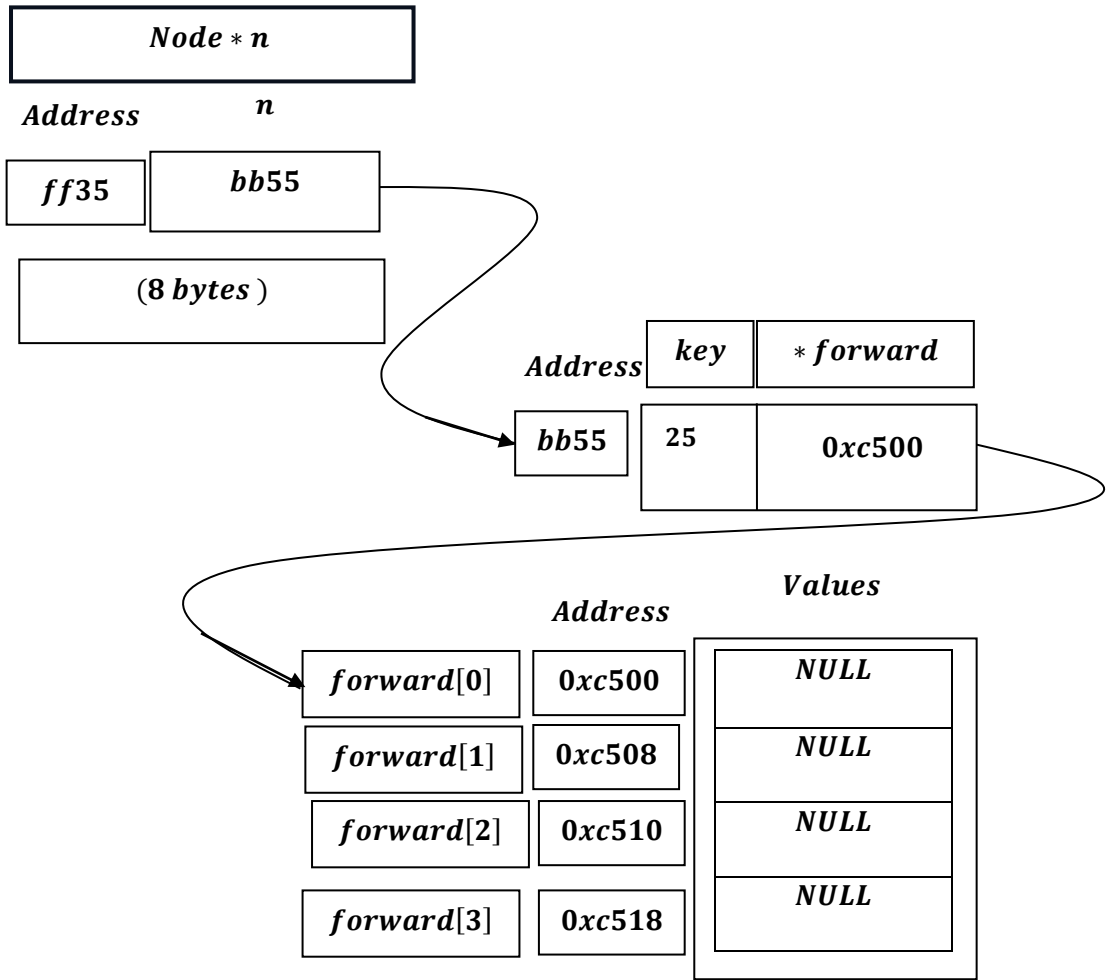
[*end of if block*]





```
struct Node *n = createNode(key, rlevel){...};
```

Let *rlevel* = 3 and *key* = 25.



```
return n;
```

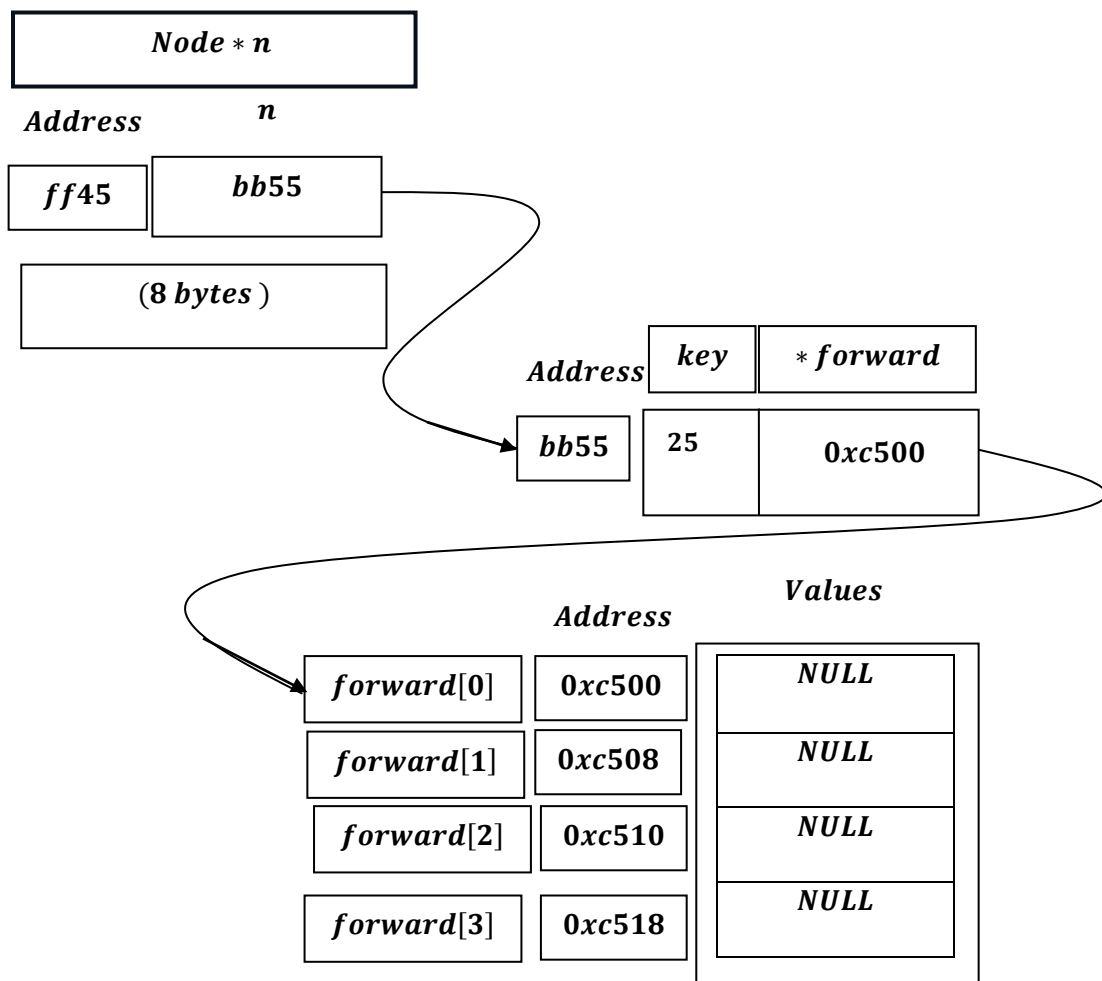
i. e. Return *n* = *bb55*

Actual structure

<i>Address at which value is stored</i>	<i>Value</i>	<i>offset used</i>
<i>bb55</i>	<i>key = 25</i>	<i>key[bb55 – bb58] 4 bytes for int</i>
		<i>padding → [bb59 – bb5c] 4 bytes for padding</i>
<i>bb5d</i>	<i>Address: 0xc500</i>	<i>* forward → [bb5d – bb64] 8 bytes for pointer variable</i>
<i>0xc500</i>	<i>NULL</i>	<i>forward[0] → [0xc500 – 0xc507] 8 bytes for pointer variable</i>
<i>0xc508</i>	<i>NULL</i>	<i>forward[1] → [0xc508 – 0xc50f] 8 bytes for pointer variable</i>
<i>0xc510</i>	<i>NULL</i>	<i>forward[2] → [0xc510 – 0xc517] 8 bytes for pointer variable</i>
<i>0xc518</i>	<i>NULL</i>	<i>forward[3] → [0xc518 – 0xc51f] 8 bytes for pointer variable</i>

struct Node * n = createNode(key = 25, rlevel = 3)

struct Node * n = bb55



```
for (int i = 0; i <= rlevel; i++) {
    n->forward[i] = update[i]->forward[i];
    update[i]->forward[i] = n;
}
```

$int\ i = 0; i \leq rlevel = 3$

$n \rightarrow forward[0] = update[0] \rightarrow forward[0] = e216$

$update[0] \rightarrow forward[0] = n = bb55$

$i = i + 1 = 0 + 1 = 1$

int i = 1; i ≤ rlevel = 3

n → forward[1] = update[1] → forward[1] = e216

update[1] → forward[1] = n = bb55

i = i + 1 = 1 + 1 = 2

int i = 2; i ≤ rlevel = 3

n → forward[2] = update[2] → forward[2] = NULL

update[2] → forward[2] = n = bb55

i = i + 1 = 2 + 1 = 3

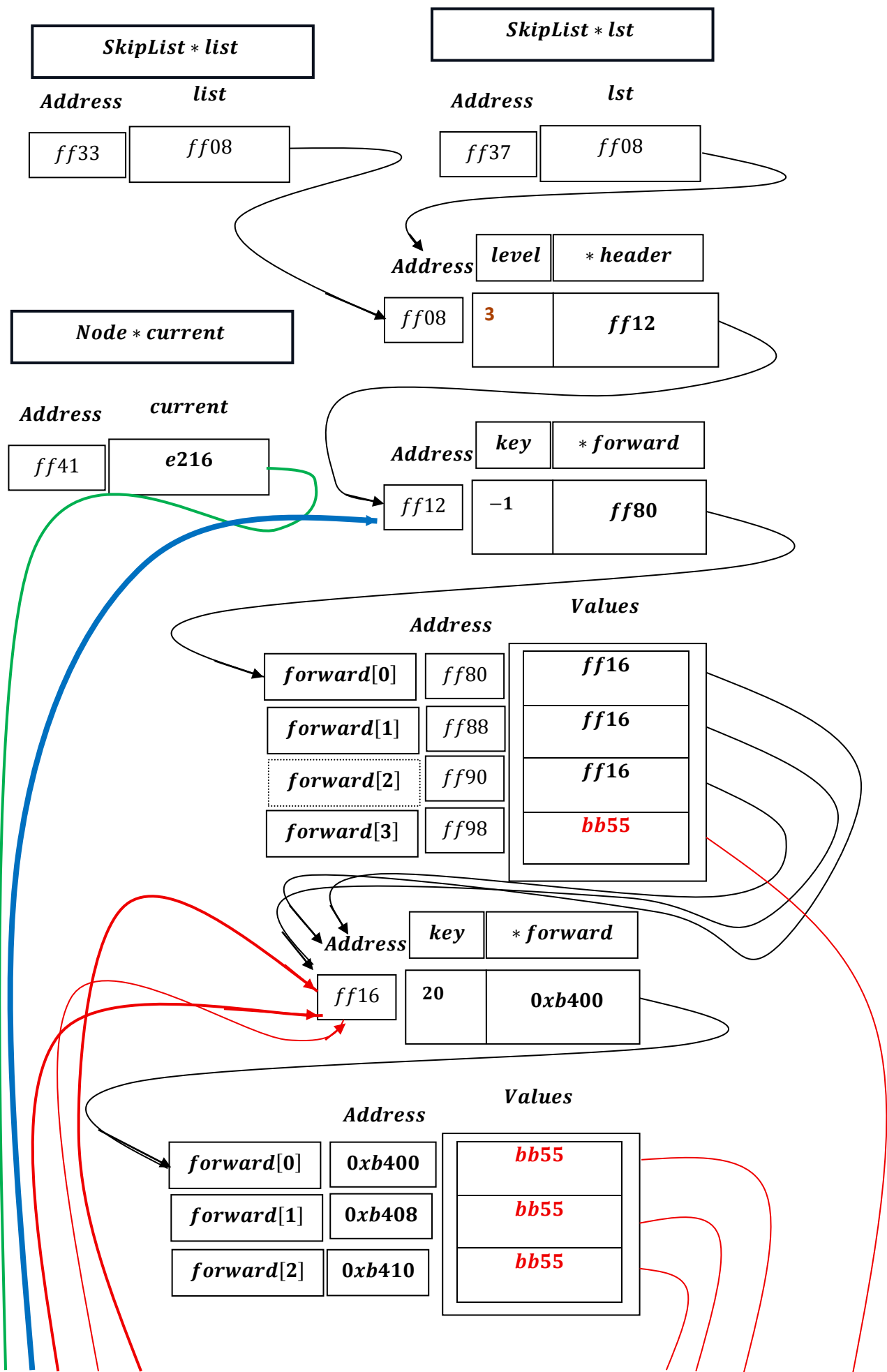
int i = 3; i ≤ rlevel = 3

n → forward[3] = update[3] → forward[3] = NULL

update[3] → forward[3] = n = bb55

i = i + 1 = 3 + 1 = 4

***int i = 4; i ≤ rlevel = 3* $\left[\begin{array}{l} \text{Condition} \\ \text{is False} \end{array} \right]$, hence loop exits**



*Node * n*

Address *n*

ff45

bb55

Address

key

** forward*

(8 bytes)

bb55

25

0xc500

Address

Values

forward[0]

0xc500

forward[1]

0xc508

forward[2]

0xc510

forward[3]

0xc518

e216

e216

NULL

NULL

Address

key

** forward*

e216

30

0xd800

Address

Values

forward[0]

0xd800

forward[1]

0xd808

NULL

NULL

Address

Values

update[0]

0x7fff0000

ff16

update[1]

0x7fff0008

ff16

update[2]

0x7fff0010

ff16

update[3]

0x7fff0018

ff12

```
cout << "Successfully inserted key " << key << endl;
```

Successfully inserted key : 25 and function ends .

As , the function ends :

*SkipList * lst*

Address	lst
ff37	ff08

*Node * current*

Address	current
ff41	e216

	Address	Values
update[0]	0x7fff0000	ff16
update[1]	0x7fff0008	ff16
update[2]	0x7fff0010	ff16
update[3]	0x7fff0018	ff12

*Node * n*

Address	n
ff45	bb55

Gets Destroyed

*SkipList * list*

Address *list*

ff33 *ff08*

Address *level* ** header*

ff08 3 *ff12*

Address *key* ** forward*

ff12 -1 *ff80*

Address

Values

<i>forward[0]</i>	<i>ff80</i>	<i>ff16</i>
<i>forward[1]</i>	<i>ff88</i>	<i>ff16</i>
<i>forward[2]</i>	<i>ff90</i>	<i>ff16</i>
<i>forward[3]</i>	<i>ff98</i>	<i>bb55</i>

Address *key* ** forward*

ff16 20 *0xb400*

Address

Values

<i>forward[0]</i>	<i>0xb400</i>	<i>bb55</i>
<i>forward[1]</i>	<i>0xb408</i>	<i>bb55</i>
<i>forward[2]</i>	<i>0xb410</i>	<i>bb55</i>

